



Biology, Geology and Environmental Sciences · 1st year of Bachillerato · Inside the Earth · Class 2 of 5 ·
With Nil

What we have touched and what we haven't



Class video: https://youtu.be/TrKzKzZ_5PM

The code leads to the video.

Name: _____ Date: _____

Before you start

1. In the previous class you saw that the Earth is twice as dense as its rocks. Where do you think the weight is?
2. **Your prediction.** What is the deepest point of the Earth's interior you think we have touched? Write down a figure in kilometres.

While you watch the video

1. Write in your own words what a direct method is and what an indirect one is.

2. Why was basalt expected in the Kola borehole at around seven kilometres? What appeared instead?
3. Why don't xenoliths sink in the magma, if they are denser than it?
4. Why can't the Earth's magnetic field be due to a magnet made of rock? Give two reasons.
5. **Your prediction.** If the temperature rose by 25 °C per kilometre all the way to the centre, how hot would it be there? What is the estimate?
6. One sixth iron (7.9 g/cm³) and five sixths rock (3.3 g/cm³): do you get the Earth's mean density, 5.5? What is missing?

The Earth in a corridor

To scale, one metre is a thousand kilometres: the Earth's radius measures 6.37 m.

What	Real depth	In the corridor
The deepest mine	about 4 km	
The Kola borehole	12.3 km	
The Spanish xenoliths	about 30–60 km	
Xenoliths and diamonds from kimberlites	150–200 km	
The deepest diamond inclusion	just over 660 km	
The edge of the core	2,891 km	
The centre	6,371 km	

What part of the corridor has no sample at all?

Practice

1. Classify as direct or indirect: a borehole, a xenolith, a gravimeter, a compass, a seismograph and a diamond with inclusions.
2. If the core takes up one sixth of the Earth's volume, its mean density is 11 and the Earth's is 5.5, what fraction of the Earth's mass is in the core?
3. What is good and what is bad about a xenolith as a sample of the mantle?

Keep thinking

Is a sample you can touch worth more than many indirect measurements that agree?

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